

# Standard Operating Procedure: Automated Procurement Workflow

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## STANDARD OPERATING PROCEDURE: AUTOMATED PROCUREMENT WORKFLOW

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### 1. PURPOSE

This SOP describes the end-to-end steps of the automated procurement workflow executed by the supply chain AI system. It is intended for operations staff who monitor the system and for auditors reviewing procurement decisions.

### 2. WORKFLOW OVERVIEW

The automated procurement workflow is triggered when the system detects that on-hand inventory for a tracked material is forecast to fall below safety stock within the next planning horizon (30 days).

Step 1 24 Inventory Check: The system retrieves current on-hand quantities and safety stock levels from the SAP inventory module.

Step 2 24 Demand Forecast: The system analyses 18 months of shipment history to produce a next-month demand forecast with confidence bounds.

Step 3 24 Procurement Proposal: If forecast demand exceeds on-hand stock, the system generates a purchase requisition proposal including recommended quantity, preferred vendor, unit price, and total estimated cost.

Step 4 24 Policy Evaluation: The proposal is evaluated against current procurement approval policies (see POL-PROC-001 and POL-PROC-002). The evaluation engine applies rules in the following priority order:

- (a) Forbidden vendor check 24 deny if vendor is on the Restricted Vendor List.
- (b) Threshold check 24 apply the approval tier matching the estimated cost.
- (c) Vendor constraint check 24 verify preferred vendor requirement is met.

Step 5 24 Outcome:

- Auto-approved: a purchase order is created automatically in SAP.
- Needs human approval: the requisition is placed in the approval queue and the responsible manager is notified.
- Denied: the workflow terminates and the operations team is notified to intervene manually.

### **3. APPROVAL QUEUE MANAGEMENT**

Approvals pending in the queue for more than 8 business hours are automatically escalated to the next management level. The operations team must review the approval queue dashboard at the start of each business day.

### **4. THRESHOLDS REFERENCE (as of SOP effective date)**

The following thresholds apply to standard catalogue materials:

- Below \$5,000 with preferred vendor: auto-approved
- \$5,000 to \$25,000: manager approval required
- Above \$25,000: executive approval required (system denies and routes to CapEx board)

These values are governed by POL-PROC-001. If POL-PROC-001 is updated, this SOP reference table must be updated in the same change cycle.

### **5. EXCEPTION HANDLING**

If the policy evaluation engine cannot extract a numeric threshold from a policy document (ambiguous language), it defaults to routing the requisition for human review. Operations staff should log such exceptions and notify the compliance team to update the policy wording.

### **6. AUDIT TRAIL**

Every automated procurement decision produces a machine-readable audit record containing: material ID, vendor ID, estimated cost, the policy rule that fired, the outcome, and a timestamp. These records are retained in the Cosmos DB audit container for a minimum of 7 years in accordance with the company records retention policy.